How Small is Small?

Cross-structural network analysis of the Small World of Singlish Words

Presentation

Data Repository

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Abstract

Psycholinguistics posit the existence of a mental lexicon that takes the shape of a network, where words are connected to each other by shared characteristics. In a word association network, nodes are words and the edges are semantic associations between words. It has been shown that semantic networks have small-world structures, where there is observed a high interconnectedness between nodes despite the large size of the network. Traditionally, these analyses of semantic association networks have been conducted through the construction of step-distance networks (SDNs). However, researchers have contended that the strength of the associations must be taken into account, which can be achieved by using an association correlation network (ACN) structure. The present study compares the SDNs and ACN of a Singlish word association dataset in order to examine its semantic structure. The results demonstrate that ACN is methodologically distinct from SDNs in representing semantic relationships, and that the semantic structure of Singlish is a small-world network across both SDN and ACN formulations.

Introduction

Using even a single word entails complex cognitive processing. Psycholinguists conduct research aimed at uncovering these processes, to create theoretical models that can explain and account for varying behavioral effects seen in word recognition processes. Popular models of language propose the existence of a mental lexicon that holds representations of words (Coltheart, 2006; Seidenberg, 2005). This mental lexicon often takes the shape of a network, where words are connected to each other by shared characteristics. One of the most notable models of cognitive psychology is one such network, known as the *spreading activation model* (Collins & Loftus, 1975). This model explains the *semantic priming* effect: the phenomenon wherein presenting a word that is semantically related to another, target word (for example, 'red') before showing the target ('fire') itself enhances the speed with which people respond to the target (See McNamara, 2005 for review).

In a spreading activation network, words represent a *node* in the network and connections or *edges* represent shared meaning between them (see Figure 1). This model posits a network of words that are not only semantically related to each other, but also elicit and enhance processing of each other. According to the spreading activation model, semantic priming follows the path of activation along edges – when we see the word 'red', its corresponding node 'red' in the mental lexicon is activated. Further, this activation of the 'red' node can elicit other words it is connected to, such as 'fire' or 'roses'. However, under this model, activation decreases along an edge across distance and accessibility – hence, the activation sent to 'fire' is proportionally more than 'roses' as it is spatially closer to 'red'. On the other hand, 'roses' will receive more activation than 'house' even though its distance from 'red' is similar – this is because the activation must first transverse the 'fire' node, which reduces the available activation to be

further sent to ‘house’.

It is apparent that, under the spreading activation model, the semantic connections between each node facilitate the priming phenomenon. The path of activation as well as the network it implies is useful to psycholinguists as it presumed to provide insight on the structure of the mental lexicon. Thus, much research has applied itself to constructing and analyzing semantic networks. In a seminal paper, Steyvers and Tenenbaum (2005) undertook an analysis of three types of semantic networks, including a word association network, to show that such networks have a *small-world structure*. In a small-world structure (Watts & Strogatz, 1998), nodes are typically clustered together into smaller groups, also known as communities. However, the presence of some well-connected nodes, which act as shortcuts, allow for efficient traversal or activation across the network. These small-world networks are characterized by their high *clustering coefficient* – the probability that the connections of a given node will themselves be connected – and small *average path length* – the average of all possible paths in the network – despite the large size of a small-world structure. In semantic networks, a small-world structure implies that small groups of words are associated together, but there are a few words which connect several other words across groups.

Traditionally, the analysis of semantic association networks as small-world structures is conducted through the construction of networks from *free association* datasets. In these datasets, participants are shown words and asked to respond with the first three words that come to mind (Nelson et al., 2004); for example, participants may be shown the word ‘red’ and responses may include ‘fire’, ‘roses’, and so on. These datasets are then formulated as *step-distance networks* (SDNs; as in Steyvers & Tenenbaum, 2005), where words are connected if one word (for example, ‘red’) evoked the response association of another word (‘fire’) by participants at least

twice. The analysis of these SDNs is assumed to shed light on the structure of the semantic mental lexicon. However, it has been extended that SDNs may not adequately display the strength of associations between words – it may not be the case that simply having an associative response between words reflect a *strong* association between words. It is thus necessary to analyse semantic networks based on the strongest associations between words. This can be achieved by assigning *weights* to the edges between the words, that denote the strength of the relation between the two nodes.

To this end, Kenett et al. (2011) put forth an *association correlation network* (ACN) model of semantic association data using a dataset in Hebrew, where the edges of the networks were correlations between the words. The dataset used was unique in that it asked participants to commit one minute to making semantic association responses, in contrast to simply answering with the top three responses (Rubinstein et al. (2005) as in Kenett et al., 2011). Kenett et al. (2011) created a correlation matrix of associations between words in this dataset using the Pearson formula. Their network is a weighted network in which the nodes are words, the edges indicate correlations between these words, and the weight indicates the strength of the correlation, resulting in a more closely-knitted network. Their analysis of the resulting ACN revealed a small-world structure of the Hebrew lexicon (see 'Hebrew ACN' column in Figure 2 for results). Further, the authors argue that their novel technique "allows us to identify how the target words organize into sub-cliques, based on semantic categories" (Kenett et al., 2011, p. 11) in a way that is distinct from the traditional SDN method.

In order to compare the SDN and ACN methodologies, Kumar et al. (2020) conducted a study that aimed to test the predictive power of each of these models by supplementing it with behavioural data. The focus of this review, however, is the construction and subsequent

comparison of both an ACN and SDN using the Nelson et al. (2004) dataset which was undertaken by the authors. It is obvious that these two types of networks capture different aspects of semantic relationships between words – indeed, it was found that the predictive power of both networks regarding semantic priming differs across task demands; however, analysis of both SDN and ACN structures of the same dataset yielded a robust small-world structure across both types of networks (Kumar et al., 2020; see Figure 2). This comparison of the SDN and ACN networks hence shows that distinct semantic relationships are captured through each of these methodologies. However, the small-world structure of the lexicon persists across both formulations.

The cross-network comparison of a semantic association dataset has only been conducted in English and Hebrew (Kenett et al., 2017; Kumar et al., 2020). However, the scope of this research can be widened considerably to accommodate a variety of languages.
Inferences derived from these networks can also be applied to networks of languages that develop from a blend of two or more formal languages, known as Creole languages (Daval-Markussen & Bakker, 2011). The present study hence aims to examine the semantic structure of Singlish words by constructing, analyzing and comparing SDN and ACN structures on the free association dataset found in Wong and Siew (2024).

Methods

Dataset

The Small World of Singlish Words (Wong & Siew, 2024) investigates the semantic associations of words unique to Singapore English, or Singlish. To create this dataset, 610 participants were presented with random words from a list of 300 Singlish words taken from Wikipedia. For each word, participants were asked to respond with the first three words that

come to their mind. If no responses can be generated, participants may choose to respond between the options ‘I do not know this word’ or ‘I recognise this word but cannot think of anything’.

Preprocessing

Each response from participants was included in this dataset, irrespective of whether the response was generated first, second or third. Populated cells from the second and third response fields were filtered using an Excel function and added to the dataset. Next, an Excel function was used to convert all responses to lowercase, in order to remove any errors that may arise from case-sensitivity. This resulted in a file with two columns of data: the first column showed the cue words, and the second column showed responses generated to the cue word (see ‘swow_responses.csv). This raw data was then imported into RStudio for further cleaning – an igraph function was used to compile the data into an edge list with weights, where the weight of each edge reflected the number of participants that generated the given response for the cue (see edge_list.csv). That is, for example, for a cue ‘fire’ and response ‘red’, a weight of 20 would indicate that 20 people responded to the word ‘fire’ with the word ‘red’.

Network Construction

Step-Distance Network (SDN). Following Steyvers and Tanenbaum (2005), only associative responses generated by at least two participants should be included in the data. Thus, any edges with a weight of 1 were deleted, leaving only edges with a weight of 2 or above. Following this, the network was converted to an unweighted network by deselecting the weight column. Following Kumar et al. (2020), a directed and an undirected SDN was constructed from the filtered edge list for further analysis.

Association-Correlation Network (ACN). The construction for the ACN in this study

differs from the Kenett et al. (2011) study; instead, it follows the method used by Kumar et al. (2020) to create a 'filtered ACN'. Here, the weighted edge list was converted to an adjacency matrix. Next, all matrix values equal to 0 were removed, and a correlation was then conducted on the resulting matrix. Thresholding was conducted as in Siew (2024), wherein the values were ordered into percentiles, and the top 30 percentile was retained for further analysis.

For both networks above, the number of nodes (n), the average number of connections (k), the average shortest path length (L), the diameter of the network (D), and the global clustering coefficient (C) were computed. For a baseline comparison, the average shortest path length and clustering coefficient for a random graph of same size and density was also computed.

Results

The results of the present study can be seen in Table 1. Similar to Kumar et al. (2020), there was high correlation between the directed and undirected SDNs. Both the directed and undirected SDN had a similar number of average connections (k = 3.19) and global clustering coefficient (C = 0.009). The directed SDN had a shorter average path length (L = 3.88) than the undirected SDN (L = 4.55), but a larger diameter (D = 10) than the undirected SDN (D = 8). Both networks showed higher clustering coefficients and lower average shortest path lengths as compared to a random network of the same size and density

In comparison to both the SDNs, the ACN showed a higher average number of connections (k = *41*.52), lower average path length (L = 1.13), smaller diameter (D = 3), and higher global clustering coefficient (C = 0.992. Similar to the SDNs, the ACN showed a higher clustering coefficient and lower average shortest path length as compared to a random network of the same size and density.

Discussion

The present study aimed to examine the semantic structure of a Singlish words using the word association dataset compiled by Wong and Siew (2024). In particular, the study aimed to achieve this by constructing directed and undirected step-distance networks (SDNs) as well as association correlation network (ACN) networks, following Steyvers and Tanenbaum (2005), Kenett et al. (2011) and Kumar et al. (2020). The analysis of these networks yielded two important results, each of which I will consider in detail below.

First, there is a higher global clustering coefficient and lower average shortest path length for each of the constructed networks as compared to a random network of the same size and density. As mentioned earlier, these parameters characterize a small-world network structure. These results follow from the analysis conducted by Kumar et al. (2020) which found the same pattern across SDNs and ACN for English word associations. This strongly affirms that, similar to formal languages, the Singlish semantic association network is a small-world network. Notably, the average shortest path length of the ACN was also shown to be smaller than that of the random network – a result that Kenett et al. (2011) did not achieve in their original study. This may imply that Singlish has more of a small-world structure than Hebrew, at least with regard to higher-level associations that characterize the ACNs under study here.

Second, the ACN and SDNs differed across network parameters, demonstrating, in line with Kenett et al. (2011), that the ACN does indeed represent a differential semantic network than the SDN. Despite being smaller than the SDNs (k = 8524), the ACN is more connected than them. The longest path length in each network, denoted by the diameter D, is much smaller for the ACN (D = 3) as compared to the directed SDN (D = 10) or the undirected SDN (D = 8). In other words, the ACN is so highly connected such that longest distance within the ACN is

significantly lesser than the SDNs. This might imply that the semantic relationships represented by the ACN can reveal a greater degree of small-world features in a network than the traditional SDNs.

To summarize, the present study has demonstrated that the semantic structure of Singlish is a small-world network across both SDN and ACN formulations, and that the ACN is methodologically distinct from SDNs in representing semantic relationships. However, there is much scope for further research in this area – this exploratory study only constructed a directed SDN, an undirected SDN, and a filtered ACN as compared to the Kumar et al. (2020) study, which compared six variations of these network structures. It would be useful to conduct an in-depth construction of PFMGs across both network types for comparison as conducted by Kumar et al. (2020) and apply it to the Singlish dataset. Moreover, these networks can then test the power of the network in predicting behavioural semantic priming results in Singlish, as the Kumar et al. (2020) study does for English.

A comparison between the Singlish network and other Creole networks may also be undertaken to uncover any unique features of these languages as compared to formal languages such as Hebrew and English. It may also be fruitful to consider results that multilingual word networks may show when applied to the SDN and ACN comparisons. In a broader lens, these semantic networks can be applied in order to address a variety of psychological questions, such as the organization of knowledge in the mind, the acquisition of language, meaning and vocabulary across development, the effects of priming and memory, and even the system of cognitive search and navigation (Siew et al., 2019).

Tables and Figures

Figure 1. A semantic association network as seen in Collins and Loftus (1975).

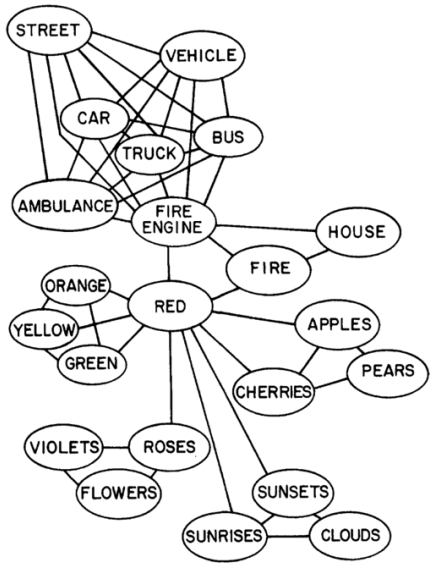


Figure 2. Comparison statistics across SDN and ACN structures in Kumar et al. (2020).

Table 1
Summary Statistics for Semantic Networks

	Step distance networks (SDNs)			Association-correlation networks (ACNs)			
Variable	Undirected SDN	Undirected PMFG	Directed SDN	ACN PMFG	Unfiltered ACN	Filtered ACN.1	Hebrew ACN
n	5018	5018	5018	5018	5018	5018	800
k	22	5.99	12.7	5.85	5018	95.19	5.94
L	3.04	12.5	4.27	23	1	2.45	10
D	5	27	10	61	1	4	25
C	0.186	0.72	0.186	0.69	1	0.26	0.68
L_{random}	3.03	1.95	4.26	1.95	3.03	—	3.94
C_{random}	0.004	0.05	0.004	0.05	0.004	—	0.005

Note. n = the number of nodes; k = the average number of connections; L = the average shortest path length; D = the diameter of the network; C = clustering coefficient; L_{random} = the average shortest path length with random graph of same size and density; C_{random} = the clustering coefficient for a random graph of same size and density; PMFG = Planar Maximally Filtered Graph.

Table 1. Summary statistics comparing the Singlish SDN and ACN in this study.

	SDN		ACN
	Directed	Undirected	
n	8705	8705	8524
k	3.19	3.19	41.52
L	3.88	4.55	1.13
D	10	8	3
C	0.009	0.009	0.992
L_random	16.27	7.81	4.05
C_random	0.0002	0.0002	0.0026

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